

SPIKE PRIME + PYTHON



**TEACHER:
CERTIFIED
TRAINER**

**DURATION:
16 CLASSES (x3 LEVELS)
(1 HOUR PER CLASS)**

**MODE:
ONLINE AND
OFFLINE**

The LEGO SPIKE™ Prime Python Curriculum is a hands-on robotics program that builds programming, logic, and engineering skills. Students progress from Python fundamentals to advanced navigation, sensor integration, and autonomous control systems. The course emphasizes problem-solving, automation, and real-world robotics applications, preparing learners for competitions and advanced STEM challenges.

COURSE CURRICULUM

Level 1: Introduction to Python Programming

- **Robotics Fundamentals:** Understanding LEGO SPIKE Prime components, motors, sensors, hub & port connections, and building stable and functional robotic models.
- **Python Programming Basics:** Writing simple Python programs, understanding program flow (top-to-bottom execution), and using print() for output and debugging.
- **Logic & Decision Making:** Conditional statements (if / elif / else), making robots react to real-world inputs, and applying logic to solve simple problems.
- **Loops & Repetition:** Using loops to repeat actions, continuous sensor checking, and understanding automation concepts.

- **Motor Control & Movement:** Controlling motor speed and direction, rotational movement and timing, coordinating multiple motors.

- **Sensor Integration:**

- Color Sensor – detecting and responding to colors
- Force Sensor – touch-based actions
- Distance Sensor – proximity-based control

- **Data Handling & Sequencing:** Using lists and dictionaries (basic introduction), sequencing steps in the correct order, mapping inputs to actions.

- **Problem Solving & Computational Thinking:** Breaking problems into smaller steps, testing, debugging, and improving logic, thinking like an engineer.

- **Creativity & Real-World Applications:** Creating music, games, and smart machines, connecting coding with real-life scenarios, encouraging innovation and imagination.

Level 2: Advanced Python Programming with SPIKE Prime

- **Advanced Driving Base:** Precise robot movement and navigation, speed control, turns, path accuracy, using multiple motors together, introduction to advanced driving logic.

- **Guitar:** Event-based interaction using sensors, mapping inputs to different sounds, creative programming with sequences, understanding real-time response systems.

- **Explorer:** Distance-based navigation, obstacle detection and avoidance, decision-making using sensor values, applying logic to unknown environments.

- **Brain Game:** Game logic and rule-based programming, counters, scoring systems and conditions, improving memory and logical thinking, designing interactive challenges.

- **Yoyo Ring:** Repetitive motion and loop mastery, timing and motor synchronization, understanding circular and oscillating motion, precision control using Python logic.

- **Break Dance:** Coordinated multi-motor movements, pattern-based motion sequences, combining loops with creative motion, exploring rhythm through robotics.

- **Smart Bike:** Real-world inspired automation concepts, speed and motion control based on input, sensor-based decision making, applying engineering logic to daily-life models.

Level 3: Advanced Line Following & Navigation

- **Zig-Zag Line Follower:** Understanding line detection using sensors, basic correction logic (left-right decisions), introducing real-time feedback, foundation of autonomous navigation.
- **Steering Line Follower:** Smooth and controlled steering movement, proportional motor speed adjustment, improving accuracy and stability, transition from simple to intelligent movement.
- **Dual Line Follower:** Using multiple sensors together, comparing sensor values simultaneously, advanced decision making, building redundancy and reliability in robots.
- **PID Line Follower:** Introduction to PID control logic, error calculation and correction, fine-tuning robot performance, industry-level robotics concepts simplified for students.
- **Gyro Straight Movement:** Using Gyro Sensor for angle control, maintaining straight-line accuracy, correcting drift and orientation errors, real-world applications in self-driving systems.

Class Structure

1. Introduction to Robotics & Programming Concepts

- Learn about LEGO SPIKE Prime components including motors, sensors, hub, and port connections.
- Understand fundamental Python concepts such as program flow, logic, and basic syntax.

2. Supervised Model Building & System Setup

- Assemble functional robotic models under trainer guidance.
- Configure motors and sensors correctly for accurate performance.

3. Understanding Programming & Control Logic

- Explore conditionals, loops, and decision-making in robotics.
- Understand how sensor data influences robot behavior in real time.
- Learn the logic behind movement control, navigation, and automation systems.

4. Practical Coding Experiments & Challenges

- Engage in structured hands-on programming tasks, including:
- Line Following Projects – Develop and optimize navigation accuracy.
- Sensor-Based Challenges – Implement obstacle detection and response systems.
- Autonomous Design Tasks – Create, test, and refine intelligent robotic solutions.